

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

– yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.

– ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.

– moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

7.03.2026. 2-smena.

1. $m = 24^3 \cdot 6^{12} \cdot 20^3$ bo'lsa, $\frac{m^2}{2^n}$ kasr qiymati natural son bo'ladigan n ning eng katta qiymatini toping.
- A) 54
B) 24
C) 27
D) 48
2. Agar $\frac{1}{2} < x < 2$ bo'lsa, $||3x - 1| - x| - |x - 2|$ ni soddalashtiring.
- A) 3
B) $3(x - 1)$
C) $x - 1$
D) 1
3. Ikki kenguru orasidagi masofa 30 m (katta kenguru kichigini quvmoqda). Katta kenguru bir sakrashda 2 m, kichigi 1 m sakraydi va katta kenguru 2 marta sakraganda kichigi uch marta sakraydi. Katta kenguru kichigini quvib yetganda katta kenguru qancha masofa o'tgan bo'ladi.
- A) 60
B) 90
C) 120
D) 180
4. Uchta ishchi 4080000 so'm ishladi. Birinchi ishchining ishlagan pulining ikkinchi ishchining ishlagan puliga nisbati $\frac{15}{2} : \frac{7}{4}$, uchinchi ishchi esa birinchi ishchi ishlagan pulining $43\frac{1}{3}\%$ qismini oladi. Shunda uchinchi ishchi qancha so'm olgan?
- A) 1060800
B) 1080600
C) 2040000
D) 2040600
5. Soddalashtiring:

$$\sqrt[3]{\frac{125 \cdot x^6}{54}}.$$

- A) $\frac{5x^{\frac{2}{5}}}{3}$
B) $\frac{5x^{\frac{1}{5}}}{3}$
C) $\frac{5x^{\frac{2}{5}}}{3\sqrt[3]{2}}$

D) $\frac{5x^{\frac{1}{5}}}{3\sqrt[3]{2}}$

6. Eng katta sonni toping.

A) 2^{222}

B) 2^{22^2}

C) $2^{2^{22}}$

D) $2^{2^{2^2}}$

7. Hisoblang:

$$\sqrt{2} \cdot (\sqrt{5} - 1) \cdot (3 + \sqrt{5}) \cdot \sqrt{3 - \sqrt{5}}.$$

A) 2

B) 8

C) 4

D) 16

8. Arifmetik progressiyaning $a_5 = 5$ va $a_7 \cdot a_{10}$ eng kichik bo'lsa, d ni toping.

A) $-1,25$

B) $-0,75$

C) $-2,25$

D) $-1,75$

9. $a_1, a_2, \dots, a_n$ arifmetik progressiya, $b_1, b_2, \dots, b_n$ geometrik progressiya hosil qiladi. $b_1 + b_2 + b_3 = 64$ va $b_1 = a_2$, $b_2 = a_4$, $b_3 = a_6$ bo'lsa, arifmetik progressiyaning n ta hadi yig'indisi 176 ga teng bo'lsa, a_n ni toping.

A) $\frac{64}{3}$

B) $\frac{32}{3}$

C) 21

D) 22

10. Soddalashtiring:

$$\frac{a^3 - b^3}{(2a + b)^2 - 3a(a + b)} + \frac{a^3 + b^3}{a^2 - ab + b^2}.$$

A) $2a - 2b$

B) $2b$

C) $2a + 2b$

D) $2a$

11. $a^3 + a - 1 = 0$ bo'lsa, $\frac{a^4 + a^3 + a^2 + 9}{a^5 - a^2 - a + 6}$ ni toping.

- A) $\frac{5}{3}$
- B) $\frac{3}{2}$
- C) 2
- D) 1

12. Hisoblang:

$$\frac{3 \sin 10^\circ - \sqrt{3} \cos 10^\circ}{3 \cos 70^\circ} .$$

- A) $-\frac{2}{\sqrt{3}}$
- B) $\frac{2}{\sqrt{3}}$
- C) $-\frac{\sqrt{3}}{2}$
- D) $\frac{\sqrt{3}}{2}$

13. x, y, z uchburchakning ichki burchaklari uchun $\sin x \cdot \sin y \cdot \sin z = \frac{1}{2}$ bo'lsa, $\sin 2x + \sin 2y + \sin 2z$ ni toping.

- A) $\frac{\sqrt{3}}{4}$
- B) $\frac{\sqrt{3}}{2}$
- C) $\frac{2\sqrt{3}}{3}$
- D) 2

14. Tenglama nechta haqiqiy ildizga ega:

$$\frac{7^x - 11^x}{\sqrt{77^x - 121^x}} = 1 .$$

- A) Haqiqiy ildizi yo'q
- B) 1 ta
- C) 2 ta
- D) 3 ta

15. Tengsizlikni yeching:

$$\sqrt{x - 11} + \sqrt{12 - x} > 0 .$$

- A) (11; 12)
- B) $(-\infty; 11) \cup (12; \infty)$
- C) {11; 12}
- D) [11; 12]

16. Tengsizlikning butun ildizlari nechta:

$$\log_{\frac{1}{2}} \left(2x - \frac{11}{2} \right) \leq \log_{4-x} 1 .$$

- A) Cheksiz ko'p yechimga ega
- B) 1 ta
- C) 2 ta
- D) Butun yechimga ega emas

17. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$\sqrt[3]{(x+3)^2} - 6\sqrt[3]{(x-3)^2} = \sqrt[3]{9-x^2} .$$

- A) $2\frac{5}{14}$
- B) $6\frac{9}{14}$
- C) 0
- D) 9

18. Tenglamaning butun ildizlari yig'indisini toping (Agar u bitta bo'lsa, o'sha ildizini toping):

$$\frac{x^2 + 3x + 46}{x^2 - 7x + 10} \leq \frac{8}{5-x} .$$

- A) -5
- B) -4
- C) -8
- D) 2

19. $x^2 - 3x + a^2 = 0$ tenglama haqiqiy ildizga ega bo'ladigan a ning nechta butun qiymati mavjud?

- A) 4 ta
- B) 1 ta
- C) 2 ta
- D) 3 ta

20. Funksiyaning aniqlanish sohasiga kiradigan butun x larni yig'indisini toping:

$$f(x) = \frac{\sqrt{x+3} + \sqrt{6-x}}{x^2 - 10x + 25} .$$

- A) 13
- B) 15
- C) 7
- D) 10

21. a ning nechta qiymatida $f(x) = x^2 + ax + 3$ va $g(x) = 2x - 1$ funksiya grafiklari bir nuqtada kesishadi?
- A) 4 ta
B) 1 ta
C) 2 ta
D) 3 ta

22. $f(x) = \frac{x + x^2 + x^3 + \dots + x^{2024}}{x + x^2 + x^3 + \dots + x^{1012}}$ bo'lsa, $f'(1)$ ni toping.

- A) 1012
B) -1012
C) 1
D) -1

23. Aniqmas integralni toping:

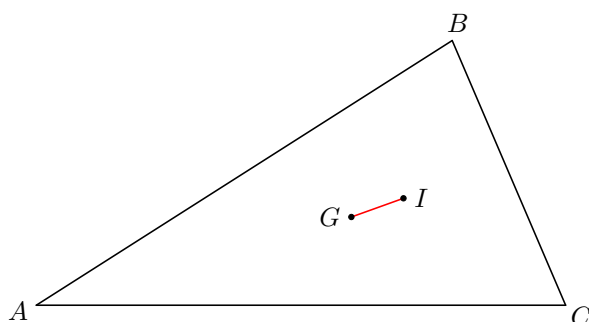
$$\int \frac{1}{x^3 + 1} dx .$$

- A) $\frac{1}{3} \ln|x+1| - \frac{1}{3} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C$
B) $\frac{1}{3} \ln|x+1| - \frac{1}{6} \ln(x^2 - x + 1) - \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C$
C) $\frac{1}{3} \ln|x+1| - \frac{1}{6} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C$
D) $\frac{1}{3} \ln|x+1| - \frac{1}{6} \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \arctan \frac{2x}{\sqrt{3}} + C$

24. Uchburchakning ikki ichki burchagi 64° va 76° ga teng. Uchburchakning bissektrisalari kesishishidan hosil bo'lgan eng kichik burchakni toping.

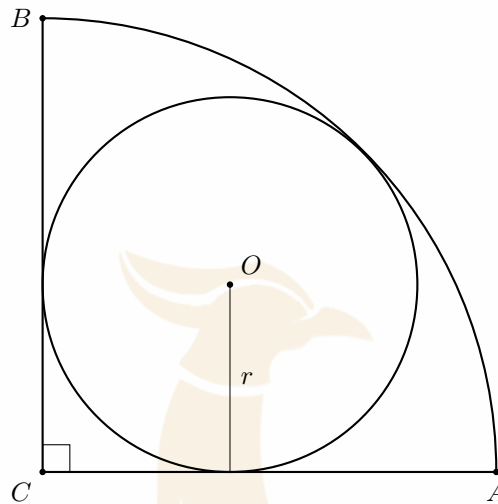
- A) 52°
B) 42°
C) 48°
D) 55°

25. ABC uchburchakning tomonlari 8, 13, 15 ga teng. Uning bissektrisalar kesishgan nuqtasidan, medianalar kesishgan nuqtasigacha bo'lgan masofani toping.



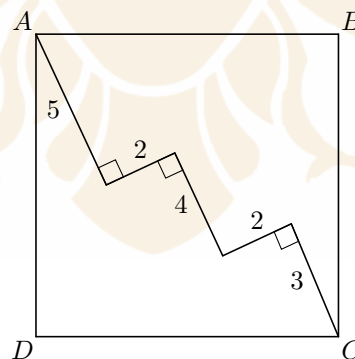
- A) $\frac{20}{3}$
 B) $\frac{\sqrt{21}}{3}$
 C) $\frac{\sqrt{19}}{3}$
 D) $\frac{\sqrt{17}}{3}$

26. Radiusi $\sqrt{2} + 1$ ga teng chorak aylana chizilgan aylana radiusini toping.



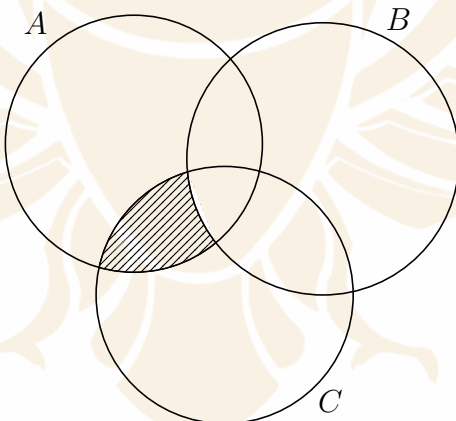
- A) $\sqrt{3}$
 B) $\sqrt{2}$
 C) $\frac{\sqrt{2}}{2}$
 D) 1

27. Chizmaga asosan kvadratni yuzini toping.



- A) 100
 B) 80
 C) 90
 D) 81

28. Muntazam oltiburchakka ichki chizilgan aylana chizilgan. Aylanaga muntazam oltiburchak ichki chizilgan. Aylanaga tashqi chizilgan oltiburchak yuzini, ichki chizilgan oltiburchak yuziga nisbatini toping.
- A) $\frac{3}{4}$
 B) $\frac{4}{5}$
 C) $\frac{4}{3}$
 D) $\frac{3}{2}$
29. $A(6; 4; -2)$, $B(x; y; 0)$ va $\overrightarrow{CD}(4; -3; 1)$ vektor berilgan. $\overrightarrow{AB}$ va $\overrightarrow{CD}$ vektorlar kollinear bo'lsa, $\overrightarrow{AB}$ ning koordinatalari yig'indisini toping.
- A) 4
 B) 12
 C) -12
 D) -4
30. Silindr o'q kesim diagonali asos radiusidan 4 marta katta. Yon sirti yuzi $12\sqrt{3}\pi$ bo'lsa, uning hajmini toping.
- A) 9π
 B) 18π
 C) 14π
 D) 12π
31. Bo'yalgan soha quyidagilardan qaysi biriga teng:



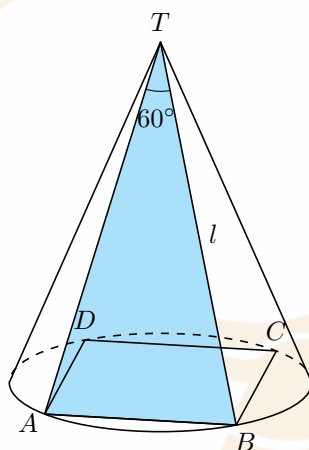
- A) $(A \cup C)/B$
 B) $B/(A \cup C)$
 C) $(A \cap C)/B$
 D) $B/(A \cap C)$
32. Nechta uch xonali sonning raqamlari yig'indisi 9 ga teng?
- A) 45
 B) 36

C) 44

D) 48

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Konus asosiga tomoni 4 ga teng bo‘lgan kvadrat ichki chizilgan. Kvadrat tomoni orqali konus uchidan kesim tushirilgan. Kesim uchidagi burchak 60° ga teng. Bu yerda $\pi \approx 3$ deb oling.

**Javoblar.**A) $2\sqrt{2}$ B) $4\sqrt{2}$ C) $2\sqrt{3}$ D) $4\sqrt{3}$ E) $16\sqrt{2}$ F) $\sqrt{2}$ **33.** Konus balandligi toping.**34.** Kesim yuzini toping.**35.** Konus hajmini toping.

36. Tengsizliklar sistemasi yeching:

$$\begin{cases} x^4 + y^4 = 17(x - y) \\ xy = 2(x - y) \end{cases}.$$

a) Tenglamalar sistemasi necha juft yechimga ega?

Javob: _____

b) Tenglamaning bir juft ildizlari yig'indisining eng kichik qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

37. Tenglamani yeching:

$$\cos 3x \sin^3 x + \sin 3x \cos^3 x = \frac{3}{4}.$$

a) Tenglamaning eng kichik musbat yechimini toping.

Javob: _____

b) Tenglamaning $(-2\pi; 2\pi)$ oralig'idagi yechimlari sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaraqasiga ko'chirib yozing.

38. $f(x) = \cos \sqrt{2}x + \cos \frac{x}{\sqrt{2}}$ funksiya berilgan.

a) Funksiyaning eng kichik musbat davrini toping.

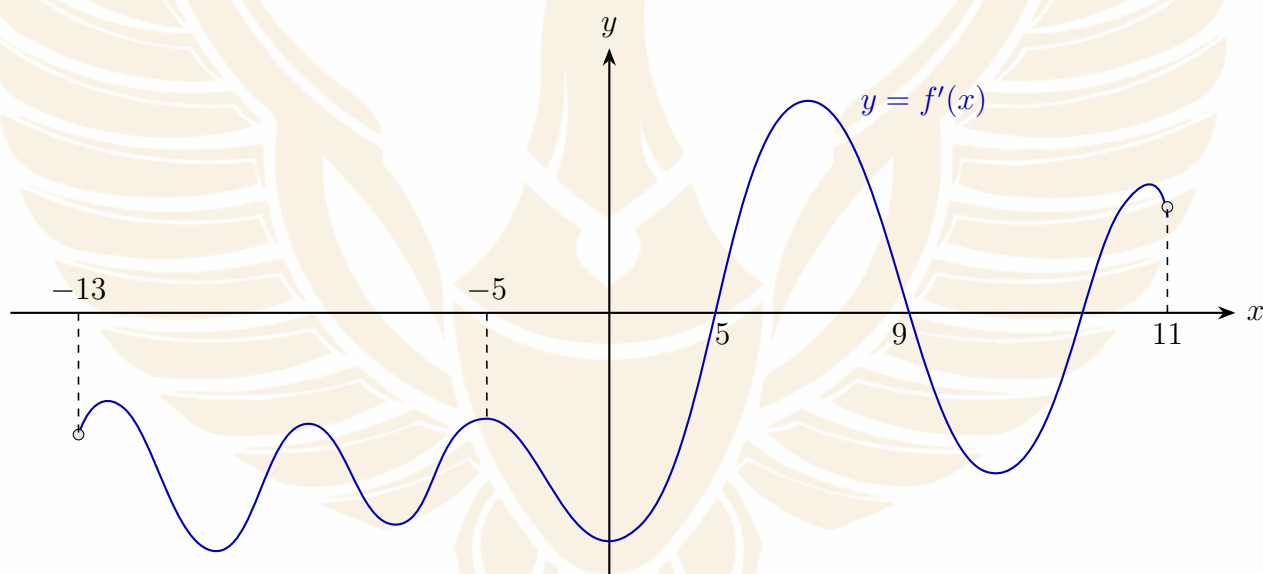
Javob: _____

b) Funksiyaning eng kichik qiymatini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

39. Quyida $f(x)$ funksiyaning hosilasi grafigi berilgan.



a) $(-5; 11)$ oralig'ida $f(x)$ funksiyaning ekstremumlari sonini toping.

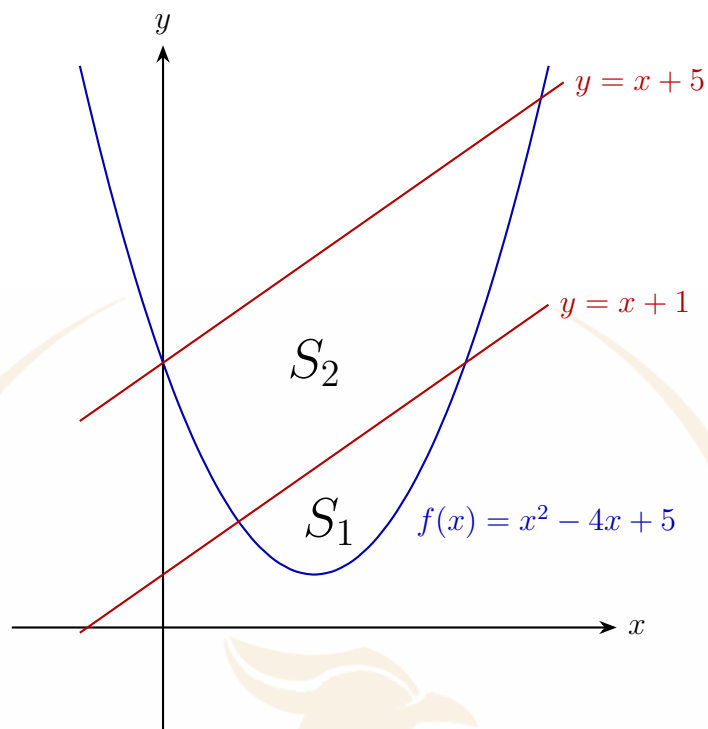
Javob: _____

b) Funksiya $(-5; 11)$ oralig'ida qaysi nuqtada maksimumga erishadi.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

40. $f(x) = x^2 - 4x + 5$, $g(x) = x + 1$, $h(x) = x + 5$ funksiyalarning grafiklari tasvirlangan.



a) S_1 ni toping.

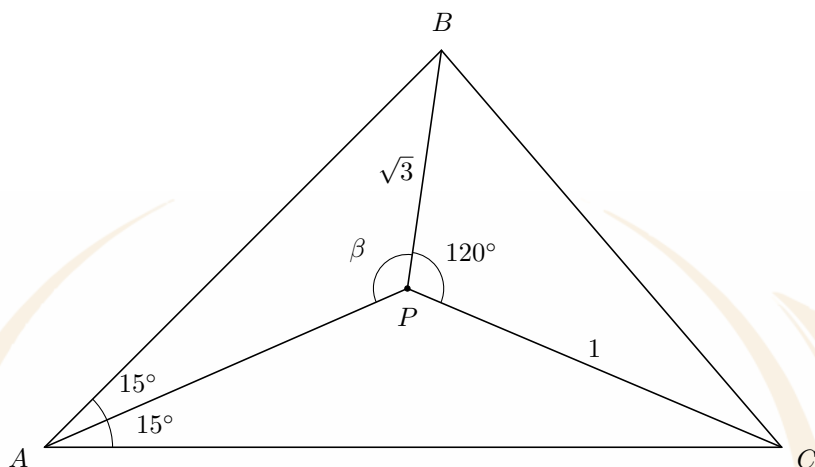
Javob: _____

b) S_2 ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. ABC uchburchakning A burchagi bissektrisasidan P nuqta olingan. $BP = \sqrt{3}$, $PC = 1$, $\angle BPC = 120^\circ$ bo'lsin.



a) $\sin \beta$ ni toping.

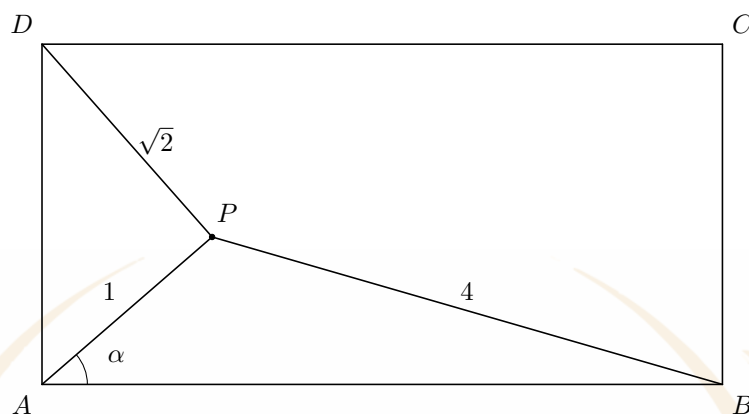
Javob: _____

b) AC tomon uzunligini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

42. $ABCD$ to'rtburchakning ichidan P nuqta olingan. $3 \cdot AD = DC$, $DP = \sqrt{2}$, $AP = 1$ va $PB = 4$ bo'lsa,



a) $\cos \alpha$ ni toping.

Javob: _____

b) To'rtburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

43. Tomoni $\sqrt{6} - \sqrt{2}$ bo'lgan muntazam o'n ikkiburchakka aylana tashqi chizilgan.

a) Aylana radiusini toping.

Javob: _____

b) O'n ikkiburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

44. Asosi rombdan iborat piramida berilgan. Rombning kichik burchagi 30° , asosiga ichki chizilgan aylana radiusi $\sqrt{3}$ ga teng. Piramida yon yoqlari asos tekisligi bilan 60° burchak hosil qiladi.

a) Piramida to'la sirti yuzini toping.

Javob: _____

b) Piramida hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

45. Beshta ishchi bir ishni bajarishmoqda. Birinchi, ikkinchi, uchinchi ishchilar birgalikda ishni 7,5 soatda, birinchi, uchinchi, beshinchi ishchilar 5 soatda, birinchi, uchinchi, to'rtinchi ishchilar 6 soatda, ikkinchi, to'rtinchi, beshinchi ishchilar 4 soatda birgalikda ishni tugatishadi.

a) Ikkinchi ishchi yolg'iz o'zi ishni necha soatda tugatadi?

Javob: _____

b) Beshta ishchi birgalikda ishni necha soatda tugatadi?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

